Convergence of Policy Gradient Methods to Nash Equilibria in Stochastic Games

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Summary

We give a self-contained, measure-theoretic treatment of the convergence of policy gradient methods to Nash equilibria in general stochastic games, following the recent work of Giannou, Lotidis, Mertikopoulos, and Vlatakis-Gkaragkounis Giannou et al. [2022]. Their main result establishes that, for any stable Nash policy in an episodic stochastic game, the sequence of policies generated by the policy gradient algorithm converges to the equilibrium with probability at least $1 - \delta$, provided the initial policy lies in a suitable neighbourhood and the step-size is sufficiently small. The proof rests on a stochastic Lyapunov argument that combines an energy inequality, martingale stability estimates, and the Robbins–Siegmund almost supermartingale convergence lemma.

Our first contribution is expository: we develop the full proof of this convergence theorem with all measure-theoretic foundations made explicit. We carefully construct the underlying probability space and filtration, verify the martingale and submartingale properties of the error processes, and state precisely the classical results from martingale theory that are invoked at each step.

Our second contribution is original: we prove that a simple random restart strategy achieves global, almost-sure convergence to a Nash equilibrium, removing the dependence on initialisation. The expected number of restarts is bounded by the inverse of the basin probability, and the framework naturally enables Pareto-optimal equilibrium selection. We validate both results empirically on normal-form and stochastic games, demonstrating that state-contingent policies strictly outperform state-blind reductions when $|\mathcal{S}| > 1$.

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Chapter 1

Introduction

1.1 Context and motivation

A stochastic game is a multi-agent extension of the Markov decision process (MDP), introduced by Shapley Shapley [1953]. At each stage of the game, the players find themselves in a common state, choose actions simultaneously, receive individual rewards, and transition to a new state according to a probability kernel that depends on the current state and the joint action profile. The players' objectives are coupled: each player's reward depends not only on her own action but on the actions of all other players.

The central solution concept in this setting is the *Nash equilibrium* Nash [1951]: a joint policy from which no player can profitably deviate. The existence of Nash equilibria in finite stochastic games was established by Fink Fink [1964], but the question of whether boundedly rational agents can *learn* to play a Nash equilibrium through repeated interaction remains largely open.

A natural approach is *policy gradient* (PG) methods: each player maintains a parameterised policy, plays the game for one episode, observes a gradient signal, and updates her policy by a gradient step Sutton and Barto [2018], Agarwal et al. [2021]. Despite the success of policy gradient methods in single-agent reinforcement learning, their convergence properties in multi-agent settings are poorly understood outside of special classes of games such as potential games or two-player zero-sum games Leonardos et al. [2022], Zhang et al. [2022], Daskalakis et al. [2020].

1.2 The main result

The paper of Giannou, Lotidis, Mertikopoulos, and Vlatakis-Gkaragkounis Giannou et al. [2022] addresses this gap. Their Theorem 1 establishes that, in a *general* finite stochastic game, the sequence of policies generated by the policy gradient algorithm converges to any *stable* Nash policy with probability at least $1 - \delta$, for any $\delta > 0$, provided the initial

policy lies in a neighbourhood of the equilibrium and the step-size is chosen appropriately. This is a local convergence guarantee: it holds conditionally on initialisation near the equilibrium, but it applies to general stochastic games without structural assumptions such as potentiality or zero-sum structure.

The result is significant for several reasons. It concerns the *actual* sequence of play, not a time-averaged or best-iterate variant. It accommodates stochastic gradient estimates with unbounded variance, as arise from the REINFORCE estimator Williams [1992]. And it provides the first trajectory convergence guarantee for policy gradient methods in this level of generality.

1.3 Contribution and outline

This dissertation presents a self-contained, measure-theoretic exposition of the proof of Theorem 1. The original proof in Giannou et al. [2022] is spread across several appendices and relies on a number of probabilistic arguments whose measure-theoretic foundations are left to the reader. Our aim is to make every step fully rigorous and explicit.

The exposition is organised as follows. Chapter 2 introduces the model: stochastic games, policies, value functions, solution concepts, and the policy gradient algorithm. We also set up the probability-theoretic framework—the underlying probability space, filtration, and the key results from martingale theory that will be needed. Chapter 3 contains the proof of the main theorem, developed through a sequence of intermediate results: an energy inequality, error aggregation via martingales and submartingales, stochastic stability, extraction of a convergent subsequence, and convergence of the energy function. Chapter 4 addresses the natural follow-up: can we remove the dependence on initialisation? We prove that a restart strategy converges to a Nash equilibrium almost surely and bound the expected number of restarts. Chapter 5 offers concluding remarks and directions for future research.

A note on attribution. The mathematical content of Chapters 2 and 3 follows Giannou et al. [2022] closely. Chapter 4 is our original contribution: it uses Theorem 3.1 as a black box to derive a global convergence guarantee via a restart argument. The measure-theoretic treatment throughout is ours.

Chapter 2

Stochastic Games and Policy Gradient Methods

The material in this chapter follows the setup of Giannou et al. [2022], to which we refer the reader for further details and references. Throughout, we use UK English conventions.

2.1 The stochastic game

We consider N-player stochastic games in the tabular, random-stopping framework. Formally, a stochastic game is specified by a tuple

$$\mathcal{G} = \big(\mathcal{S},\ \mathcal{N},\ \{\mathcal{A}_i, R_i\}_{i\in\mathcal{N}},\ P,\ \zeta,\ \rho\big), \tag{2.1}$$

whose components are defined as follows.

- $\mathcal{N} = \{1, 2, \dots, N\}$ is a finite set of *agents*.
- $\mathcal{S} = \{1, 2, \dots, S\}$ is a finite set of *states*.
- For each agent $i \in \mathcal{N}$, the set $\mathcal{A}_i = \{1, \dots, A_i\}$ is a finite set of *actions*. We write $\mathcal{A} = \prod_{i\in\mathcal{N}} \mathcal{A}_i$ for the joint action space, and $\mathcal{A}_{-i} = \prod_{j\neq i} \mathcal{A}_j$ for the action space of all agents other than i.
- For each $i \in \mathcal{N}$, the map $R_i \colon \mathcal{S} \times \mathcal{A} \to [-1, 1]$ is the *reward function* of agent i.
- $P(s' \mid s, \alpha)$ denotes the probability of transitioning from state s to state s' when the joint action profile is $\alpha \in \mathcal{A}$.
- The game terminates at each state–action pair (s, α) with probability $\zeta_{s,\alpha} = 1 - \sum_{s'\in\mathcal{S}} P(s' \mid s, \alpha) > 0$. We write $\zeta := \min_{s,\alpha} \zeta_{s,\alpha}$ for the minimum termination probability.
- $\rho \in \Delta(\mathcal{S})$ is the distribution of the initial state.

Episodic setting. Play proceeds in *episodes*. In each episode, the initial state s_0 is drawn from ρ. At each time step $t \geqslant 0$, all agents observe the common state s_t, select actions $\alpha_t \in \mathcal{A}$, and receive rewards $\{R_i(s_t, \alpha_t)\}_{i \in \mathcal{N}}$. The game then terminates with probability ζ_{s_t, α_t}, or transitions to s_{t+1} drawn from $P(\cdot \mid s_t, \alpha_t)$. We denote the resulting trajectory by $\tau = (s_t, \alpha_t, r_t)_{t \leqslant T(\tau)}$, where $T(\tau)$ is the (random) termination time.

2.2 Policies and value functions

We consider *stationary Markovian* policies: maps from states to distributions over actions that do not depend on time or history. A *policy* for agent i is a map $\pi_i \colon \mathcal{S} \to \Delta(\mathcal{A}_i)$, which we view as an element of the set $\Pi_i \coloneqq \Delta(\mathcal{A}_i)^S$. A *policy profile* is a tuple $\pi = (\pi_i)_{i \in \mathcal{N}} \in \Pi \coloneqq \prod_{i \in \mathcal{N}} \Pi_i$. We write λ_Π for the Lebesgue measure on Π (the product of the Lebesgue measures on each simplex Π_i).

The *value function* of agent i under policy profile π, starting from state s, is

$$V_{i,s}(\pi) \coloneqq \mathbb{E}_{\tau \sim \mathrm{MDP}}\left[\sum_{t=0}^{T(\tau)} R_i(s_t, \alpha_t) \;\middle|\; s_0 = s\right], \tag{2.2}$$

where the expectation is over the randomness induced by π and the transition kernel P. Setting $V_{i,\rho}(\pi) \coloneqq \mathbb{E}_{s \sim \rho}[V_{i,s}(\pi)]$, we obtain the *expected value* under initial distribution ρ.

Discounted state visitation distribution. For a policy profile $\pi \in \Pi$ and initial distribution ρ, define the *unnormalised discounted state visitation measure*

$$\bar{d}_\rho^\pi(s) = \mathbb{E}_{\tau \sim \mathrm{MDP}}\left[\sum_{t=0}^{T(\tau)} \mathbb{1}\{s_t = s\} \;\middle|\; s_0 \sim \rho\right], \tag{2.3}$$

and the normalised version $d_\rho^\pi(s) = \bar{d}_\rho^\pi(s)/Z_\rho^\pi$, where $Z_\rho^\pi = \sum_{s \in \mathcal{S}} \bar{d}_\rho^\pi(s)$. Since the termination probability satisfies $\zeta > 0$, we have $\bar{d}_\rho^\pi(s) \leqslant 1/\zeta < \infty$ for all s, so these quantities are well-defined.

The mismatch coefficient. The quantity

$$C_{\mathcal{G}} \coloneqq \max_{\pi, \pi' \in \Pi} \left\{ \left\| \bar{d}_\rho^\pi / \bar{d}_\rho^{\pi'} \right\|_\infty \right\} \tag{2.4}$$

measures the worst-case ratio of visitation distributions across policy profiles. Following Agarwal et al. [2021], Giannou et al. [2022], we assume $C_{\mathcal{G}} < \infty$, which is equivalent to $d_\rho^\pi(s) > 0$ for all π and s.

2.3 Nash policies and their refinements

Definition 2.1 (Nash policy)**.** A policy $\pi^* = (\pi_i^*)_{i\in\mathcal{N}} \in \Pi$ is a *Nash policy* for the initial distribution ρ if, for every player $i \in \mathcal{N}$ and every unilateral deviation $\pi_i \in \Delta(\mathcal{A}_i)^S$,

$$V_{i,\rho}(\pi_i^*;\, \pi_{-i}^*) \geqslant V_{i,\rho}(\pi_i;\, \pi_{-i}^*). \tag{NE}$$

We write $v_i(\pi) = \nabla_i V_{i,\rho}(\pi)$ for the gradient of player i's value function with respect to π_i (keeping π_{-i} fixed), and $v(\pi) = (v_i(\pi))_{i\in\mathcal{N}}$ for the concatenated gradient vector.

A key structural property of stochastic games, established in Giannou et al. [2022], Daskalakis et al. [2020], Zhang et al. [2022], is the following.

Lemma 2.2 (Gradient dominance property; [Giannou et al., 2022, Lemma 2])**.** *For any policy profile $\pi = (\pi_i)_{i\in\mathcal{N}} \in \Pi$ and any unilateral deviation $\pi_i' \in \Pi_i$ of player i,*

$$V_{i,\rho}(\pi_i';\, \pi_{-i}) - V_{i,\rho}(\pi_i;\, \pi_{-i}) \leqslant C_{\mathcal{G}} \max_{\bar{\pi}_i\in\Pi_i} \langle \nabla_i V_{i,\rho}(\pi),\, \bar{\pi}_i - \pi_i\rangle. \tag{GDP}$$

An immediate consequence is that first-order stationary points of V are Nash [Giannou et al., 2022, Lemma 3].

Lemma 2.3 (First-order stationarity implies Nash)**.** *A policy $\pi^* \in \Pi$ is Nash if and only if it satisfies the first-order stationary condition*

$$\langle v(\pi^*),\, \pi - \pi^*\rangle \leqslant 0 \quad \textit{for all } \pi \in \Pi. \tag{FOS}$$

We now introduce refinements of Nash policies that play a central role in the convergence analysis.

Definition 2.4 (Refinements of Nash policies; [Giannou et al., 2022, Definition 2])**.** Let $\pi^* \in \Pi$ be a Nash policy.

- π^* is *stable* if $\langle v(\pi),\, \pi - \pi^*\rangle < 0$ for all $\pi \neq \pi^*$ sufficiently close to π^*.
- π^* is *second-order stationary* (SOS) if

$$(\pi - \pi^*)^\top \operatorname{Jac}_v(\pi^*)(\pi - \pi^*) < 0 \quad \text{for all } \pi \in \Pi \setminus \{\pi^*\}, \tag{SOS}$$

  where $\operatorname{Jac}_v(\pi^*) = (\nabla_j v_i(\pi^*))_{i,j\in\mathcal{N}}$ is the Jacobian of v at π^*.
- π^* is *deterministic* if $\pi_i^*(s)$ places all mass on a single action for every i and s.
- π^* is *strict* if it is deterministic and (FOS) holds with strict inequality whenever $\pi \neq \pi^*$.

One has the implications strict/deterministic $\Longrightarrow$ SOS $\Longrightarrow$ stable $\Longrightarrow$ FOS = Nash; see [Giannou et al., 2022, Proposition 1]. The following quantitative versions of the stability condition will be used in the proof.

Proposition 2.5 (Drift inequalities; [Giannou et al., 2022, Proposition 1]). *Let $\pi^* \in \Pi$ be a Nash policy.*

(a) *If π^* is second-order stationary, there exists $\mu > 0$ such that*

$$\langle v(\pi),\, \pi - \pi^* \rangle \leqslant -\mu \, \|\pi - \pi^*\|^2 \quad \textit{for all } \pi \textit{ sufficiently close to } \pi^*. \tag{2.5}$$

(b) *If π^* is strict, there exists $\mu > 0$ such that*

$$\langle v(\pi),\, \pi - \pi^* \rangle \leqslant -\mu \, \|\pi - \pi^*\| \quad \textit{for all } \pi \textit{ sufficiently close to } \pi^*. \tag{2.6}$$

2.4 The policy gradient algorithm

We now describe the inter-episode learning framework. At the beginning of each episode $n = 1, 2, \dots$, each agent i commits to a policy $\pi_{i,n} \in \Pi_i$. The agents play the game under the joint policy $\pi_n = (\pi_{i,n})_{i \in \mathcal{N}}$, observe a gradient signal $\hat{v}_n$, and update their policies according to the *policy gradient* (PG) template:

$$\pi_{n+1} = \operatorname{proj}_{\Pi}(\pi_n + \gamma_n \hat{v}_n), \tag{PG}$$

where $\operatorname{proj}_{\Pi} \colon \prod_i \mathbb{R}^{A_i \times S} \to \Pi$ denotes the Euclidean projection onto the feasible policy space, and $\gamma_n > 0$ is the step-size. We take the step-size to be of the form

$$\gamma_n = \frac{\gamma}{(n+m)^p}, \tag{2.7}$$

for some $\gamma > 0$, $m \geqslant 0$, and $p \in (1/2, 1]$.

Gradient signal decomposition. The gradient signal $\hat{v}_n$ is decomposed as

$$\hat{v}_n = v(\pi_n) + U_n + b_n, \tag{2.8}$$

where

$$U_n := \hat{v}_n - \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n], \qquad b_n := \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n] - v(\pi_n). \tag{2.9}$$

Here $\mathcal{F}_n$ is the natural filtration defined in Section 2.5 below. The term U_n is a zero-mean noise (a martingale difference) and b_n is a systematic bias. We assume deterministic upper bounds:

$$\mathbb{E}\big[\|b_n\| \,\big|\, \mathcal{F}_n\big] \leqslant B_n, \qquad \mathbb{E}\big[\|U_n\|^2 \,\big|\, \mathcal{F}_n\big] \leqslant \sigma_n^2, \tag{2.10}$$

where $B_n = \mathcal{O}(1/n^{\ell_b})$ and $\sigma_n = \mathcal{O}(n^{\ell_\sigma})$ for some $\ell_b, \ell_\sigma > 0$ depending on the information model.

REINFORCE estimator. A concrete instance of this framework arises from the REINFORCE algorithm Williams [1992]. Each agent plays the game with a *sampling policy* $\tilde{\pi}_{i,n}$ (possibly different from $\pi_{i,n}$) and constructs the gradient estimate

$$\hat{v}_i = R_i(\tau) \cdot \sum_{t=0}^{T(\tau)} \nabla_i \big(\log \tilde{\pi}_i(\alpha_{i,t} \mid s_t)\big). \tag{2.11}$$

In the ε-Greedy PG variant, one sets $\tilde{\pi}_{i,n} = (1 - \varepsilon_n)\pi_{i,n} + \varepsilon_n \operatorname{Unif}_{\mathcal{A}_i}$. This introduces bias of order $B_n = O(\varepsilon_n)$ and bounds the variance at $\sigma_n^2 = O(1/\varepsilon_n)$; see [Giannou et al., 2022, Lemma 4].

2.5 Probability-theoretic framework

We now set up the measure-theoretic foundations that underpin the convergence analysis. This section is not present in Giannou et al. [2022]; it makes explicit the structure that is used implicitly throughout their proofs.

The probability space. We work on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ rich enough to support the entire learning process: the initial policy π_1, all episode trajectories $\tau_1, \tau_2, \ldots$, and hence all subsequent policies $\pi_2, \pi_3, \ldots$ determined by (PG). Formally, one may construct $(\Omega, \mathcal{F}, \mathbb{P})$ as the product of the law of π_1 with a countable product of copies of the trajectory space, using the Ionescu-Tulcea extension theorem Durrett [2019]; the precise construction is standard and we omit the details.

Filtration. The natural filtration for this process is

$$\mathcal{F}_n := \sigma\big(\pi_1, \hat{v}_1, \hat{v}_2, \ldots, \hat{v}_{n-1}\big) \quad \text{for } n \geqslant 1, \tag{2.12}$$

with $\mathcal{F}_1 = \sigma(\pi_1)$. This σ-algebra contains all information available *before* episode n is played. In particular:

- π_n is $\mathcal{F}_n$-measurable, since π_n is a deterministic function of π_{n-1} and $\hat{v}_{n-1}$ via (PG).
- $\hat{v}_n$ is *not* $\mathcal{F}_n$-measurable: it depends on the trajectory τ_n, which is generated *during* episode n.
- $\hat{v}_n$ is $\mathcal{F}_{n+1}$-measurable by definition.

Since π_n is $\mathcal{F}_n$-measurable and v is a deterministic function, $v(\pi_n)$ is $\mathcal{F}_n$-measurable. The bias $b_n = \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n] - v(\pi_n)$ is likewise $\mathcal{F}_n$-measurable. The noise U_n and all quantities derived from it (such as ξ_n, defined in Chapter 3) are $\mathcal{F}_{n+1}$-measurable.

Conditional independence between episodes. The key structural property is that each episode trajectory τ_n is conditionally independent of all previous trajectories given the

policy π_n. More precisely, for any bounded measurable function f,

$$\mathbb{E}\big[f(\tau_n) \,\big|\, \mathcal{F}_n\big] = \mathbb{E}\big[f(\tau_n) \,\big|\, \pi_n\big] \quad \text{a.s.} \tag{2.13}$$

This follows from the Markov property of the episodic game: given π_n, the distribution of τ_n is completely determined by the MDP dynamics and does not depend on previous trajectories. It is this property that ensures $\mathbb{E}[U_n \mid \mathcal{F}_n] = 0$.

Classical results from martingale theory. We shall invoke the following standard results. We state them here for easy reference; proofs can be found in Williams [1991], Durrett [2019], Hall and Heyde [1980].

Theorem 2.6 (Doob's maximal inequality; [Durrett, 2019, Theorem 4.4.2])**.** *Let $(X_n)_{n\geqslant 0}$ be a nonnegative submartingale. Then for any $\lambda > 0$,*

$$\mathbb{P}\Big(\max_{0\leqslant k\leqslant n} X_k \geqslant \lambda\Big) \leqslant \frac{\mathbb{E}[X_n]}{\lambda}. \tag{2.14}$$

Theorem 2.7 (Strong law for martingale differences; [Hall and Heyde, 1980, Theorem 2.18])**.** *Let $(M_n)_{n\geqslant 1}$ be a square-integrable martingale with differences $d_n = M_n - M_{n-1}$. If $c_n \to \infty$ and*

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[d_n^2]}{c_n^2} < \infty, \tag{2.15}$$

then $M_n/c_n \to 0$ almost surely.

Theorem 2.8 (Doob's submartingale convergence; [Durrett, 2019, Theorem 4.2.11])**.** *If $(X_n)_{n\geqslant 1}$ is a submartingale satisfying $\sup_n \mathbb{E}[X_n^+] < \infty$, then X_n converges almost surely to a finite limit.*

Theorem 2.9 (Robbins–Siegmund lemma; Robbins and Siegmund [1971], [Polyak, 1987, p. 49])**.** *Let $(\mathcal{F}_n)$ be a filtration and let (X_n), (a_n), (b_n) be nonnegative adapted sequences satisfying*

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leqslant X_n + a_n - b_n \quad \textit{a.s. for all } n, \tag{2.16}$$

with $\sum_{n=1}^{\infty} a_n < \infty$ a.s. Then X_n converges almost surely to a finite limit and $\sum_{n=1}^{\infty} b_n < \infty$ a.s.

Notation. We write $\|\Pi\| := \max_{\pi,\pi'\in\Pi} \|\pi - \pi'\|$ for the diameter of the policy space, and $G := \sup_\pi \|v(\pi)\|$ for the gradient bound; both are finite since Π is compact and v is continuous [Giannou et al., 2022, Lemmas D.5, D.7].

Chapter 3

Asymptotic Convergence to Stable Nash Policies

This chapter contains the proof of the main convergence result. The exposition follows Appendix A of Giannou et al. [2022], with all measure-theoretic details made explicit.

3.1 Statement of the main theorem

Theorem 3.1 (Convergence to stable Nash policies; [Giannou et al., 2022, Theorem 1])**.** *Let π^* be a stable Nash policy, and let π_n be the sequence of play generated by* (PG) *with step-size $\gamma_n = \gamma/(n+m)^p$, $p \in (1/2, 1]$, and policy gradient estimates such that $p + \ell_b > 1$ and $p - \ell_\sigma > 1/2$, as per* (2.10)*. Then there exists a neighbourhood $\mathcal{U}$ of π^* in Π such that, for any given $\delta > 0$, we have*

$$\mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}) \geqslant 1 - \delta, \tag{3.1}$$

provided that γ is small enough (or m large enough) relative to δ.

3.2 Overview of the proof strategy

The proof proceeds in five steps, which we summarise before carrying them out in detail.

1. **Energy inequality** (Section 3.3). We show that the squared distance $D(\pi) = \frac{1}{2}\|\pi - \pi^*\|^2$ acts as a local Lyapunov function: $D(\pi_n)$ decreases on average near π^*, up to error terms arising from the noise and bias.
2. **Error aggregation** (Section 3.4). We aggregate the error terms into a martingale M_n (zero-mean errors) and a submartingale S_n (bias and variance), and study their sum $W_n = M_n^2 + S_n$.
3. **Stochastic stability** (Section 3.5). We show that, with probability at least $1 - \delta$, the iterates π_n remain within a ball $\mathcal{B}$ centred at π^* for all n. The key tool is Doob's

maximal inequality, applied to bound the probability of "bad realisations" of the noise.

4. **Convergent subsequence** (Section 3.6). Using the strong law of large numbers for martingale differences and Doob's convergence theorem, we extract a subsequence $\pi_{n_k} \to \pi^*$ with probability one on the stability event.
5. **Convergence of the energy** (Section 3.7). We show that $D_n = D(\pi_n)$ converges to a finite limit using the Robbins–Siegmund lemma. Combined with the existence of a convergent subsequence, this forces $D_n \to 0$, i.e., $\pi_n \to \pi^*$.

3.3 The energy inequality

Define the *energy function*

$$D(\pi) := \tfrac{1}{2}\|\pi - \pi^*\|^2, \tag{3.2}$$

and set $D_n := D(\pi_n)$ for the energy along the iterates.

Lemma 3.2 (Energy inequality; [Giannou et al., 2022, Lemma A.1])**.** *For all $n \geqslant 1$,*

$$D_{n+1} \leqslant D_n + \gamma_n \langle v(\pi_n),\, \pi_n - \pi^* \rangle + \gamma_n \xi_n + \gamma_n \chi_n + \gamma_n^2 \psi_n^2, \tag{3.3}$$

where the error terms are

$$\xi_n = \langle U_n,\, \pi_n - \pi^* \rangle, \qquad \chi_n = \|\Pi\|\, B_n, \qquad \psi_n^2 = \tfrac{1}{2}\|\hat{v}_n\|^2. \tag{3.4}$$

Proof. By the definition of the iterates (PG) and the fact that $\pi^* \in \Pi$,

$$D_{n+1} = \tfrac{1}{2}\|\pi_{n+1} - \pi^*\|^2 = \tfrac{1}{2}\|\operatorname{proj}_\Pi(\pi_n + \gamma_n \hat{v}_n) - \operatorname{proj}_\Pi(\pi^*)\|^2. \tag{3.5}$$

Since the Euclidean projection onto a closed convex set is non-expansive (i.e., $\|\operatorname{proj}_\Pi(x) - \operatorname{proj}_\Pi(y)\| \leqslant \|x - y\|$ for all x, y), and $\operatorname{proj}_\Pi(\pi^*) = \pi^*$, we obtain

$$D_{n+1} \leqslant \tfrac{1}{2}\|\pi_n + \gamma_n \hat{v}_n - \pi^*\|^2. \tag{3.6}$$

Expanding the squared norm gives

$$\begin{aligned} D_{n+1} &\leqslant \tfrac{1}{2}\|\pi_n - \pi^*\|^2 + \gamma_n \langle \hat{v}_n,\, \pi_n - \pi^* \rangle + \tfrac{1}{2}\gamma_n^2 \|\hat{v}_n\|^2 \\ &= D_n + \gamma_n \langle \hat{v}_n,\, \pi_n - \pi^* \rangle + \gamma_n^2 \psi_n^2. \end{aligned} \tag{3.7}$$

Substituting the decomposition $\hat{v}_n = v(\pi_n) + U_n + b_n$ from (2.8), the inner product becomes

$$\langle \hat{v}_n,\, \pi_n - \pi^* \rangle = \langle v(\pi_n),\, \pi_n - \pi^* \rangle + \underbrace{\langle U_n,\, \pi_n - \pi^* \rangle}_{=\xi_n} + \langle b_n,\, \pi_n - \pi^* \rangle. \tag{3.8}$$

For the bias term, the Cauchy–Schwarz inequality yields

$$\langle b_n,\, \pi_n - \pi^* \rangle \leqslant \|b_n\| \cdot \|\pi_n - \pi^*\| \leqslant \|b_n\| \cdot \|\Pi\| \leqslant B_n \cdot \|\Pi\| = \chi_n, \tag{3.9}$$

where in the last inequality we used the conditional bound $\|b_n\| \leqslant B_n$ that holds with probability one (since B_n is a deterministic upper bound on $\mathbb{E}[\|b_n\| \mid \mathcal{F}_n]$, and we bound $\|b_n\|$ deterministically by the diameter of Π; more precisely, the bound $\langle b_n, \pi_n - \pi^* \rangle \leqslant \|\Pi\| B_n$ follows from taking the expectation of $\langle b_n, \pi_n - \pi^* \rangle$ conditional on $\mathcal{F}_n$ and using Cauchy–Schwarz: $\mathbb{E}[\langle b_n, \pi_n - \pi^* \rangle \mid \mathcal{F}_n] \leqslant \|\pi_n - \pi^*\| \mathbb{E}[\|b_n\| \mid \mathcal{F}_n] \leqslant \|\Pi\| B_n$). Combining (3.7)–(3.9) gives (3.3). □

Remark 3.3. The terms in (3.3) play distinct roles. The drift term $\gamma_n \langle v(\pi_n), \pi_n - \pi^* \rangle$ is *negative* near π^* by the stability condition, driving the energy downward. The noise term $\gamma_n \xi_n$ has zero conditional mean and will be controlled via martingale theory. The bias term $\gamma_n \chi_n$ and the variance term $\gamma_n^2 \psi_n^2$ are nonnegative errors that must be shown to be summable.

3.4 Error aggregation

We now aggregate the errors appearing in (3.3). Define, for $n \geqslant 1$,

$$M_n := \sum_{k=1}^{n} \gamma_k \xi_k, \qquad S_n := \sum_{k=1}^{n} \left[\gamma_k \chi_k + \gamma_k^2 \psi_k^2 \right]. \tag{3.10}$$

Lemma 3.4 (Martingale and submartingale properties)**.** *With respect to the filtration $(\mathcal{F}_{n+1})_{n \geqslant 0}$:*

(a) *$(M_n)_{n \geqslant 1}$ is a square-integrable martingale.*

(b) *$(S_n)_{n \geqslant 1}$ is a submartingale. In fact, S_n is nondecreasing, since each summand is nonnegative.*

Proof. Adaptedness. Since $\xi_k = \langle U_k, \pi_k - \pi^* \rangle$ and U_k is $\mathcal{F}_{k+1}$-measurable (see Section 2.5), each summand $\gamma_k \xi_k$ is $\mathcal{F}_{k+1}$-measurable. Thus M_n is $\mathcal{F}_{n+1}$-measurable, i.e., (M_n) is adapted to $(\mathcal{F}_{n+1})_{n \geqslant 0}$. The same argument applies to S_n.

Integrability of M_n. We have $|\xi_k| \leqslant \|U_k\| \cdot \|\Pi\|$, so $\mathbb{E}[\xi_k^2] \leqslant \|\Pi\|^2 \mathbb{E}[\|U_k\|^2] \leqslant \|\Pi\|^2 \sigma_k^2 < \infty$. Since $M_n = \sum_{k=1}^{n} \gamma_k \xi_k$ is a finite sum of square-integrable random variables, M_n is square-integrable.

Martingale property. It suffices to show $\mathbb{E}[\gamma_n \xi_n \mid \mathcal{F}_n] = 0$. Since γ_n is deterministic and $\pi_n - \pi^*$ is $\mathcal{F}_n$-measurable, we have

$$\mathbb{E}[\gamma_n \xi_n \mid \mathcal{F}_n] = \gamma_n \langle \mathbb{E}[U_n \mid \mathcal{F}_n], \pi_n - \pi^* \rangle = 0, \tag{3.11}$$

where we used $\mathbb{E}[U_n \mid \mathcal{F}_n] = 0$ (which follows from $U_n = \hat{v}_n - \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n]$ by definition) and the fact that a $\mathcal{F}_n$-measurable factor can be pulled out of a conditional expectation on $\mathcal{F}_n$. By the tower property, $\mathbb{E}[M_n \mid \mathcal{F}_n] = M_{n-1} + \mathbb{E}[\gamma_n \xi_n \mid \mathcal{F}_n] = M_{n-1}$.

Submartingale property of S_n. Each increment $\gamma_k \chi_k + \gamma_k^2 \psi_k^2 \geqslant 0$ (since χ_k, ψ_k^2 are nonneg), so $S_n \geqslant S_{n-1}$ pointwise. □

Following Hsieh et al. [2019], Giannou et al. [2022], we introduce the *mean square error process*

$$W_n := M_n^2 + S_n. \tag{3.12}$$

Lemma 3.5. *The process $(W_n)_{n\geqslant 1}$ is a submartingale with respect to $(\mathcal{F}_{n+1})$.*

Proof. Since M_n is a martingale, (M_n^2) is a submartingale by Jensen's inequality applied to the convex function $x \mapsto x^2$: $\mathbb{E}[M_n^2 \mid \mathcal{F}_n] \geqslant (\mathbb{E}[M_n \mid \mathcal{F}_n])^2 = M_{n-1}^2$. Since S_n is also a submartingale, their sum $W_n = M_n^2 + S_n$ is a submartingale. □

3.5 Error control and stochastic stability

Let $\mathcal{B} = \{\pi \in \Pi : \|\pi - \pi^*\| \leqslant \varrho\}$ be a ball of radius ϱ centred at π^* in Π, chosen so that $\langle v(\pi), \pi - \pi^*\rangle < 0$ for all $\pi \in \mathcal{B} \setminus \{\pi^*\}$ (this is possible by the stability of π^*; without loss of generality, we take $\mathcal{B}$ maximal in this regard).

Fix a parameter $a > 0$ satisfying $2a + \sqrt{a} < \varrho$, and define the neighbourhood

$$\mathcal{U} := \{\pi \in \Pi : D(\pi) \leqslant a\} = \left\{\pi \in \Pi : \|\pi - \pi^*\|^2/2 \leqslant a\right\}. \tag{3.13}$$

We assume $\pi_1 \in \mathcal{U}$, i.e., the process is initialised in a ball of radius $\sqrt{2a}$ around π^*.

Define the "good" events

$$\mathcal{E}_n := \{\pi_k \in \mathcal{B} \text{ for all } k = 1, \dots, n\}, \qquad H_n := \{W_k \leqslant a \text{ for all } k = 1, \dots, n\}, \tag{3.14}$$

with the convention $\mathcal{E}_0 = H_0 = \Omega$.

The following technical lemma provides the recursive estimates needed for stability. Its proof is the most involved part of the argument.

Lemma 3.6 (Error control; [Giannou et al., 2022, Lemma A.2])**.** *Let π_n be the sequence generated by* (PG) *with $\pi_1 \in \mathcal{U}$. Then:*

(a) *$\mathcal{E}_{n+1} \subseteq \mathcal{E}_n$ and $H_{n+1} \subseteq H_n$ for all $n \geqslant 1$.*

(b) *$H_{n-1} \subseteq \mathcal{E}_n$ for all $n \geqslant 1$.*

(c) *Consider the "bad realisation" event*

$$\tilde{H}_n := H_{n-1} \setminus H_n = \{W_k \leqslant a \text{ for } k = 1, \dots, n-1 \text{ and } W_n > a\}, \tag{3.15}$$

and set $\tilde{W}_n := W_n \mathbb{1}_{H_{n-1}}$. Then

$$\mathbb{E}[\tilde{W}_n] \leqslant \mathbb{E}[\tilde{W}_{n-1}] + \gamma_n \|\Pi\| B_n + \gamma_n^2 \|\Pi\|^2 \sigma_n^2 + \tfrac{3}{2}\gamma_n^2 (G^2 + B_n^2 + \sigma_n^2) - a\,\mathbb{P}(\tilde{H}_{n-1}). \tag{3.16}$$

Proof. Part (a) is immediate from the definitions: both $\mathcal{E}_n$ and H_n are defined as intersections of events up to time n, so adding one more constraint can only shrink them.

For part (b), assume H_{n-1} holds, so $W_k \leqslant a$ for all $k = 1, \dots, n-1$. By telescoping the energy inequality (3.3) and using $\langle v(\pi_k), \pi_k - \pi^* \rangle \leqslant 0$ on $\mathcal{E}_k$ (which holds when $\pi_k \in \mathcal{B}$), we obtain, for any $k \leqslant n$,

$$D_{k+1} \leqslant D_1 + M_k + S_k \leqslant a + \sqrt{W_k} \leqslant a + \sqrt{a}, \tag{3.17}$$

where we used $D_1 \leqslant a$ and $W_k = M_k^2 + S_k \geqslant M_k^2 \geqslant 0$, so $|M_k| + S_k \leqslant |M_k| + M_k^2 + S_k - M_k^2 + S_k \leqslant \sqrt{W_k} + W_k$; in fact, the key step is the bound from the energy inequality. Specifically, telescoping (3.3) over $k = 1, \dots, n$ and dropping the negative drift term gives

$$D_{n+1} \leqslant D_1 + M_n + S_n. \tag{3.18}$$

On H_n, we have $W_n = M_n^2 + S_n \leqslant a$. Since $M_n^2 \geqslant 0$ and $S_n \geqslant 0$, this implies $|M_n| \leqslant \sqrt{a}$ and $S_n \leqslant a$. Therefore $D_{n+1} \leqslant a + \sqrt{a} + a = 2a + \sqrt{a} < \varrho^2/2$, so $\pi_{n+1} \in \mathcal{B}$, confirming $\mathcal{E}_{n+1}$. By induction, $H_{n-1} \subseteq \mathcal{E}_n$.

Part (c) requires a detailed computation. We write $W_n - W_{n-1}$ explicitly using (3.12):

$$\begin{aligned} W_n &= (M_{n-1} + \gamma_n \xi_n)^2 + S_{n-1} + \gamma_n \chi_n + \gamma_n^2 \psi_n^2 \\ &= W_{n-1} + 2M_{n-1}\gamma_n \xi_n + \gamma_n^2 \xi_n^2 + \gamma_n \chi_n + \gamma_n^2 \psi_n^2. \end{aligned} \tag{3.19}$$

Taking conditional expectations given $\mathcal{F}_n$ (and noting that M_{n-1} and $\mathbb{1}_{H_{n-1}}$ are $\mathcal{F}_n$-measurable, since they depend only on $\hat{v}_1, \dots, \hat{v}_{n-1}$ and π_1):

$$\begin{aligned} \mathbb{E}[W_n \mid \mathcal{F}_n] &= W_{n-1} + 2M_{n-1}\gamma_n \underbrace{\mathbb{E}[\xi_n \mid \mathcal{F}_n]}_{=0} + \gamma_n^2 \mathbb{E}[\xi_n^2 \mid \mathcal{F}_n] + \gamma_n \chi_n + \gamma_n^2 \mathbb{E}[\psi_n^2 \mid \mathcal{F}_n] \\ &\geqslant W_{n-1}, \end{aligned} \tag{3.20}$$

confirming that W_n is a submartingale.

Now define $\tilde{W}_n = W_n \mathbb{1}_{H_{n-1}}$ and decompose:

$$\begin{aligned} \tilde{W}_n &= W_n \mathbb{1}_{H_{n-1}} \\ &= W_{n-1}\mathbb{1}_{H_{n-1}} + (W_n - W_{n-1})\mathbb{1}_{H_{n-1}} \\ &= W_{n-1}\mathbb{1}_{H_{n-2}} - W_{n-1}\mathbb{1}_{\tilde{H}_{n-1}} + (W_n - W_{n-1})\mathbb{1}_{H_{n-1}}, \end{aligned} \tag{3.21}$$

where we used $H_{n-1} = H_{n-2} \setminus \tilde{H}_{n-1}$, so $\mathbb{1}_{H_{n-1}} = \mathbb{1}_{H_{n-2}} - \mathbb{1}_{\tilde{H}_{n-1}}$. Thus

$$\tilde{W}_n = \tilde{W}_{n-1} + (W_n - W_{n-1})\mathbb{1}_{H_{n-1}} - W_{n-1}\mathbb{1}_{\tilde{H}_{n-1}}. \tag{3.22}$$

Taking expectations, we bound each piece:

The noise term. Since H_{n-1} and M_{n-1} are both $\mathcal{F}_n$-measurable, the tower property gives

$$\mathbb{E}[M_{n-1}\gamma_n \xi_n \mathbb{1}_{H_{n-1}}] = \mathbb{E}\big[M_{n-1}\mathbb{1}_{H_{n-1}} \gamma_n \mathbb{E}[\xi_n \mid \mathcal{F}_n]\big] = 0. \tag{3.23}$$

The squared noise. Using $\xi_n^2 = |\langle U_n, \pi_n - \pi^* \rangle|^2 \leqslant \|U_n\|^2 \|\pi_n - \pi^*\|^2 \leqslant \|\Pi\|^2 \|U_n\|^2$ and $\mathbb{E}[\|U_n\|^2 \mid \mathcal{F}_n] \leqslant \sigma_n^2$:

$$\mathbb{E}[\xi_n^2 \mathbb{1}_{H_{n-1}}] \leqslant \mathbb{E}\big[\mathbb{1}_{H_{n-1}} \|\Pi\|^2 \mathbb{E}[\|U_n\|^2 \mid \mathcal{F}_n]\big] \leqslant \|\Pi\|^2 \sigma_n^2. \tag{3.24}$$

The variance term. Since $\|\hat{v}_n\| \leqslant \|v(\pi_n)\| + \|U_n\| + \|b_n\|$ and $\|v(\pi_n)\| \leqslant G$, we have $\mathbb{E}[\psi_n^2 \mathbb{1}_{H_{n-1}}] = \mathbb{E}[\frac{1}{2}\|\hat{v}_n\|^2 \mathbb{1}_{H_{n-1}}] \leqslant \frac{3}{2}(G^2 + B_n^2 + \sigma_n^2)$.

The bias term. $\mathbb{E}[\chi_n \mathbb{1}_{H_{n-1}}] \leqslant \|\Pi\| B_n$.

The penalty. On $\tilde{H}_{n-1}$, we have $W_{n-1} > a$ (since $\tilde{H}_{n-1} = H_{n-2} \setminus H_{n-1}$ means $W_{n-1} > a$ while $W_k \leqslant a$ for $k < n-1$), so $\mathbb{E}[W_{n-1} \mathbb{1}_{\tilde{H}_{n-1}}] \geqslant a\, \mathbb{P}(\tilde{H}_{n-1})$.

Combining these estimates with (3.22) yields (3.16). □

We now establish the main stability result.

Proposition 3.7 (Stochastic stability; [Giannou et al., 2022, Proposition A.1])**.** *Fix $\delta > 0$ and let π_n be generated by (PG) with step-size and gradient estimates as in Theorem 3.1. Then*

$$\mathbb{P}(H_n \mid \pi_1 \in \mathcal{U}) \geqslant 1 - \delta \quad \text{for all } n = 1, 2, \dots, \tag{3.25}$$

provided γ is sufficiently small (or m sufficiently large) relative to δ.

Proof. We bound $\mathbb{P}(\tilde{H}_n)$ using the "bad realisation" event (3.15). If $\pi_1 \in \mathcal{U}$, then on $\tilde{H}_n$ we have $W_n > a$ and $W_k \leqslant a$ for all $k < n$. Since $W_n \geqslant 0$, we have $\mathbb{1}_{\tilde{H}_n} \leqslant W_n \mathbb{1}_{H_{n-1}}/a$, so

$$\mathbb{P}(\tilde{H}_n) = \mathbb{E}[\mathbb{1}_{\tilde{H}_n}] \leqslant \mathbb{E}[W_n \mathbb{1}_{H_{n-1}}]/a = \mathbb{E}[\tilde{W}_n]/a. \tag{3.26}$$

Telescoping (3.16) from 1 to n gives

$$\mathbb{E}[\tilde{W}_n] \leqslant \mathbb{E}[\tilde{W}_0] + \|\Pi\| \sum_{k=1}^{n} \gamma_k B_k + \sum_{k=1}^{n} \gamma_k^2 \varrho_k^2 - a \sum_{k=1}^{n} \mathbb{P}(\tilde{H}_{k-1}), \tag{3.27}$$

where $\varrho_k^2 := \|\Pi\|^2 \sigma_k^2 + \frac{3}{2}(G^2 + B_k^2 + \sigma_k^2)$ and $\tilde{W}_0 = W_0 = 0$ by convention. Therefore

$$\sum_{k=1}^{n} \mathbb{P}(\tilde{H}_k) \leqslant \frac{1}{a} \sum_{k=1}^{n} \left[\gamma_k B_k \|\Pi\| + \gamma_k^2 \varrho_k^2\right] =: \frac{C}{a}, \tag{3.28}$$

where $C \equiv C(\gamma, m) > 0$ satisfies $\lim_{\gamma \to 0^+} C(\gamma, m) = 0$ and $\lim_{m \to \infty} C(\gamma, m) = 0$, by the summability conditions on γ_n, B_n, σ_n (since $p + \ell_b > 1$ and $2p - 2\ell_\sigma > 1$).

By choosing γ small enough (or m large enough), we can ensure $C/a < \delta$. Since the events $\tilde{H}_k$ are disjoint (each $\tilde{H}_k$ specifies the *first* time W exceeds a),

$$\mathbb{P}\Big(\bigcup_{k=1}^{n} \tilde{H}_k\Big) = \sum_{k=1}^{n} \mathbb{P}(\tilde{H}_k) \leqslant \delta. \tag{3.29}$$

Since $H_n = \left(\bigcup_{k=1}^{n} \tilde{H}_k\right)^c \cap \{W_0 \leqslant a\}$, we conclude $\mathbb{P}(H_n \mid \pi_1 \in \mathcal{U}) \geqslant 1 - \delta$. □

3.6 Extraction of a convergent subsequence

Proposition 3.8 (Convergent subsequence; [Giannou et al., 2022, Proposition A.2])**.** *Let π^* be a stable Nash policy. Under the assumptions of Theorem 3.1, the sequence π_n admits a subsequence π_{n_k} converging to π^* with probability 1 on the event $\mathcal{E} := \bigcap_n \mathcal{E}_n = \{\pi_n \in \mathcal{B} \text{ for all } n\}$.*

Proof. Define $\mathcal{Q} := \{\pi_n \in \mathcal{B} \text{ for all } n\} \cap \{\liminf_n \|\pi_n - \pi^*\| > 0\}$ — the event that π_n stays in $\mathcal{B}$ but does *not* admit a subsequence converging to π^*. We show $\mathbb{P}(\mathcal{Q}) = 0$.

Assume for contradiction that $\mathbb{P}(\mathcal{Q}) > 0$. On $\mathcal{Q}$, the stability of π^* gives a (random but positive) constant $c > 0$ such that $\langle v(\pi_n), \pi_n - \pi^* \rangle \leqslant -c$ for all n. Summing the energy inequality (3.3) from $k = 1$ to n and using $D_n \geqslant 0$:

$$0 \leqslant D_{n+1} \leqslant D_1 - c\,\tau_n + M_n + S_n, \tag{3.30}$$

where $\tau_n = \sum_{k=1}^n \gamma_k$. Since $p \leqslant 1$, we have $\tau_n \to \infty$. Dividing by τ_n:

$$0 \leqslant \frac{D_1}{\tau_n} - c + \frac{M_n}{\tau_n} + \frac{S_n}{\tau_n}. \tag{3.31}$$

$M_n/\tau_n \to 0$ *a.s.* The martingale differences $d_n = \gamma_n \xi_n$ satisfy

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[d_n^2]}{\tau_n^2} \leqslant \sum_{n=1}^{\infty} \frac{\gamma_n^2 \|\Pi\|^2 \sigma_n^2}{\tau_n^2}. \tag{3.32}$$

Since $\gamma_n^2 \sigma_n^2 = O(\gamma_n^2 \cdot n^{2\ell_\sigma})$ and $p - \ell_\sigma > 1/2$ implies $\sum \gamma_n^2 \sigma_n^2 < \infty$ (as $2p - 2\ell_\sigma > 1$), and $\tau_n \to \infty$, the strong law for martingale differences (Theorem 2.7) gives $M_n/\tau_n \to 0$ a.s.

$S_n/\tau_n \to 0$ *a.s.* By the bounds on B_n and σ_n,

$$\mathbb{E}[S_n] = \sum_{k=1}^{n} \left[\gamma_k \chi_k + \gamma_k^2 \mathbb{E}[\psi_k^2]\right] \leqslant \|\Pi\| \sum_{k=1}^{n} \gamma_k B_k + \tfrac{3}{2} \sum_{k=1}^{n} \gamma_k^2 (G^2 + B_k^2 + \sigma_k^2). \tag{3.33}$$

Since $p + \ell_b > 1$, the first sum converges, and since $2p - 2\ell_\sigma > 1$, the second converges as well. Hence S_n is bounded in L^1, and by Doob's submartingale convergence theorem (Theorem 2.8), $S_n \to S_\infty$ a.s. for some finite S_∞. In particular, $S_n/\tau_n \to 0$.

Letting $n \to \infty$ in (3.31), $0 \leqslant -c + 0 + 0 = -c < 0$, a contradiction. Since $D \geqslant 0$ and $\mathbb{P}(\mathcal{Q}) > 0$ was assumed, we conclude $\mathbb{P}(\mathcal{Q}) = 0$. □

3.7 Convergence of the energy function

Proposition 3.9 (Energy convergence; [Giannou et al., 2022, Proposition A.3])**.** *Under the assumptions of Proposition 3.7, there exists a finite random variable D_∞ such that*

$$\mathbb{P}\big(D_n \to D_\infty \text{ as } n \to \infty \mid \pi_n \in \mathcal{B} \text{ for all } n\big) = 1. \tag{3.34}$$

Proof. Let $\mathcal{E}_n = \{\pi_k \in \mathcal{B} \text{ for all } k = 1, \dots, n\}$ as before, and define $\tilde{D}_n := \mathbb{1}_{\mathcal{E}_n} D_n$. Since $\mathcal{E}_{n+1} \subseteq \mathcal{E}_n$ (Lemma 3.6(a)), we have $\mathbb{1}_{\mathcal{E}_{n+1}} \leqslant \mathbb{1}_{\mathcal{E}_n}$, and so

$$\tilde{D}_{n+1} = \mathbb{1}_{\mathcal{E}_{n+1}} D_{n+1} \leqslant \mathbb{1}_{\mathcal{E}_n} D_{n+1}. \tag{3.35}$$

On $\mathcal{E}_n$, the energy inequality (3.3) combined with the stability condition $\langle v(\pi_k), \pi_k - \pi^* \rangle \leqslant 0$ gives

$$\mathbb{1}_{\mathcal{E}_n} D_{n+1} \leqslant \mathbb{1}_{\mathcal{E}_n} D_n + \gamma_n \mathbb{1}_{\mathcal{E}_n} \xi_n + \mathbb{1}_{\mathcal{E}_n} \big(\gamma_n \chi_n + \gamma_n^2 \psi_n^2\big). \tag{3.36}$$

Thus $\tilde{D}_{n+1} \leqslant \tilde{D}_n + \gamma_n \mathbb{1}_{\mathcal{E}_n} \xi_n + \gamma_n \chi_n + \gamma_n^2 \psi_n^2$. Taking conditional expectations given $\mathcal{F}_n$:

$$\begin{aligned}\mathbb{E}[\tilde{D}_{n+1} \mid \mathcal{F}_n] &\leqslant \tilde{D}_n + \gamma_n \mathbb{1}_{\mathcal{E}_n} \underbrace{\mathbb{E}[\xi_n \mid \mathcal{F}_n]}_{=0} + \gamma_n \chi_n + \gamma_n^2 \mathbb{E}[\psi_n^2 \mid \mathcal{F}_n] \\ &\leqslant \tilde{D}_n + \gamma_n \|\Pi\| B_n + \tfrac{3}{2}\gamma_n^2 (G^2 + B_n^2 + \sigma_n^2). \end{aligned} \tag{3.37}$$

Denoting the nonnegative remainder by $a_n := \gamma_n \|\Pi\| B_n + \frac{3}{2}\gamma_n^2 (G^2 + B_n^2 + \sigma_n^2)$, we have $\sum_n a_n < \infty$ by the step-size and bias/variance assumptions. The Robbins–Siegmund lemma (Theorem 2.9), applied with $X_n = \tilde{D}_n$, $b_n = 0$, and a_n as above, yields that $\tilde{D}_n$ converges almost surely to a finite random variable $\tilde{D}_\infty$.

On the event $\mathcal{E} = \bigcap_n \mathcal{E}_n$, we have $\mathbb{1}_{\mathcal{E}_n} = 1$ for all n, so $\tilde{D}_n = D_n$ and hence $D_n \to \tilde{D}_\infty$ a.s. on $\mathcal{E}$. Setting $D_\infty = \tilde{D}_\infty$ on $\mathcal{E}$ gives the result. □

3.8 Proof of the main theorem

We are now in a position to prove Theorem 3.1.

Proof of Theorem 3.1. Let $\mathcal{E} = \bigcap_n \mathcal{E}_n = \{\pi_n \in \mathcal{B} \text{ for all } n\}$. By Proposition 3.7 (via Lemma 3.6(b), which gives $H_n \subseteq \mathcal{E}_{n+1}$), we have

$$\mathbb{P}(\mathcal{E} \mid \pi_1 \in \mathcal{U}) \geqslant 1 - \delta, \tag{3.38}$$

noting that the neighbourhood $\mathcal{U}$ is independent of the confidence level δ.

By Propositions 3.8 and 3.9, the following hold with probability 1 on $\mathcal{E}$:

(i) $\liminf_n \|\pi_n - \pi^*\| = 0$ (there is a subsequence converging to π^*).

(ii) D_n converges to a finite limit D_∞.

Since $D_n = \frac{1}{2}\|\pi_n - \pi^*\|^2$, statement (i) implies that some subsequence $D_{n_k} \to 0$. Statement (ii) says $D_n \to D_\infty$. A convergent sequence with a subsequence converging to zero must itself converge to zero: hence $D_\infty = 0$ and $\pi_n \to \pi^*$.

Therefore

$$\begin{aligned}\mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}) &\geqslant \mathbb{P}(\mathcal{E} \cap \{\pi_n \to \pi^*\} \mid \pi_1 \in \mathcal{U}) \\ &= \mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}, \mathcal{E})\, \mathbb{P}(\mathcal{E} \mid \pi_1 \in \mathcal{U}) \\ &\geqslant 1 \cdot (1 - \delta) = 1 - \delta, \end{aligned} \tag{3.39}$$

completing the proof. □

3.9 Application to the REINFORCE algorithm

We conclude this chapter by recording the consequence of Theorem 3.1 for the three gradient information models described in Section 2.4.

Corollary 3.10 ([Giannou et al., 2022, Corollary 1]). *Suppose that* (PG) *is run with step-size* $\gamma_n = \gamma/(n+m)^p$, $p > 1/2$, *and, if applicable, an exploration parameter* $\varepsilon_n = \varepsilon/(n+m)^r$ *with* $1 - p < r < 2p - 1$. *Then:*

- Full gradients *(*$\hat{v}_n = v(\pi_n)$*, so* $B_n = 0$, $\sigma_n = 0$*): the conclusions of Theorem 3.1 hold for any* $p > 1/2$.
- Stochastic gradients *(*$B_n = 0$, $\sigma_n = O(1)$*): same conclusion, with* $p > 1/2$.
- ε-Greedy REINFORCE *(*$B_n = O(\varepsilon_n)$, $\sigma_n = O(1/\sqrt{\varepsilon_n})$*): the conclusion holds for* $p > 2/3$.

Proof. For full and stochastic gradients, $\ell_b = \infty$ and $\ell_\sigma = 0$, so the conditions $p + \ell_b > 1$ and $p - \ell_\sigma > 1/2$ reduce to $p > 1/2$.

For the ε-Greedy variant, $B_n = O(\varepsilon_n)$ with $\varepsilon_n = \varepsilon/(n+m)^r$, giving $\ell_b = r$. The variance satisfies $\sigma_n = O(1/\sqrt{\varepsilon_n}) = O(n^{r/2})$, giving $\ell_\sigma = r/2$. The conditions become $p + r > 1$ and $p - r/2 > 1/2$, i.e., $r > 1 - p$ and $r < 2p - 1$. For such an r to exist, we need $1 - p < 2p - 1$, i.e., $p > 2/3$. □

Chapter 4

Global Convergence via Random Restarts

Theorem 3.1 guarantees convergence to a stable Nash policy provided the initial policy lies in a neighbourhood $\mathcal{U}$ of the equilibrium. In practice, agents do not know where Nash policies are located, so the dependence on initialisation is a serious limitation. This chapter removes it: we prove that a simple restart strategy converges to a Nash equilibrium *almost surely*, regardless of how the game is structured, provided at least one stable Nash policy exists.

4.1 The restart policy gradient algorithm

The idea is to run independent PG trajectories from fresh random initialisations. If any trajectory starts inside the basin of attraction, Theorem 3.1 ensures it converges.

Definition 4.1 (Restart-PG)**.** Fix step-size parameters γ, m, p satisfying the conditions of Theorem 3.1. A *restart schedule* is a sequence of independent runs $(R_k)_{k\geqslant 1}$, where each run R_k consists of:

1. An initialisation $\pi_1^{(k)}$ drawn independently from a distribution ν on Π.
2. A PG trajectory $(\pi_n^{(k)})_{n\geqslant 1}$ generated by (PG) from $\pi_1^{(k)}$, using independent episode trajectories.

We say epoch k is *successful* if $\pi_n^{(k)}$ converges to some stable Nash policy as $n \to \infty$.

4.2 Product probability space

The measure-theoretic foundation for the restart argument is a product construction. Each epoch k is supported on a probability space $(\Omega_k, \mathcal{F}_k, \mathbb{P}_k)$ as constructed in Section 2.5,

carrying the initialisation $\pi_1^{(k)}$ and all episode trajectories $\tau_1^{(k)}, \tau_2^{(k)}, \ldots$ of that epoch. The full restart process lives on the product space

$$(\Omega, \mathcal{F}, \mathbb{P}) = \prod_{k=1}^{\infty} (\Omega_k, \mathcal{F}_k, \mathbb{P}_k), \tag{4.1}$$

constructed via the Kolmogorov extension theorem Durrett [2019]. Since each factor is independent, the success events

$$A_k := \left\{\omega \in \Omega : \pi_n^{(k)}(\omega) \to \pi^* \text{ for some stable Nash } \pi^*\right\} \tag{4.2}$$

are mutually independent: A_k depends only on the k-th coordinate of the product space.

4.3 Almost-sure convergence

We make the following assumptions.

(A1) The game $\mathcal{G}$ possesses at least one strict Nash policy (Definition 2.4). Since every strict Nash policy is stable, the local convergence guarantee of Theorem 3.1 applies at each such equilibrium.

(A2) The initialisation distribution ν is absolutely continuous with respect to the Lebesgue measure λ_Π on Π, with Radon–Nikodym density bounded below: $d\nu/d\lambda_\Pi \geqslant c$ for some $c > 0$. (The uniform distribution on Π satisfies this with $c = 1/\lambda_\Pi(\Pi)$.)

Let $\pi^{*,1}, \ldots, \pi^{*,J}$ denote the stable Nash policies of $\mathcal{G}$, with corresponding neighbourhoods $\mathcal{U}_1, \ldots, \mathcal{U}_J \subseteq \Pi$ from Theorem 3.1. Define

$$p := \nu\Big(\bigcup_{j=1}^{J} \mathcal{U}_j\Big) > 0, \tag{4.3}$$

which is strictly positive because each $\mathcal{U}_j$ is a ball of positive radius in Π and ν has full support.

Theorem 4.2 (Global convergence via restarts)**.** *Under assumptions (A1) and (A2), choose step-size parameters γ, m so that Theorem 3.1 holds with some confidence level $\delta \in (0, 1)$. Then:*

(a) *The restart schedule converges to a Nash equilibrium almost surely:*

$$\mathbb{P}\big(\exists\, k \geqslant 1 : A_k \text{ occurs}\big) = 1. \tag{4.4}$$

(b) *The index K^* of the first successful epoch satisfies*

$$\mathbb{P}(K^* > k) \leqslant \big(1 - p(1-\delta)\big)^k \quad \text{for all } k \geqslant 1, \tag{4.5}$$

and in particular $\mathbb{E}[K^*] \leqslant 1/\big(p(1-\delta)\big)$.

Proof. By Theorem 3.1, for each epoch k,

$$\mathbb{P}(A_k) \geqslant \mathbb{P}\big(\pi_1^{(k)} \in \textstyle\bigcup_j \mathcal{U}_j\big) \cdot \min_j \mathbb{P}\big(\pi_n^{(k)} \to \pi^{*,j} \,\big|\, \pi_1^{(k)} \in \mathcal{U}_j\big) \geqslant p(1-\delta). \tag{4.6}$$

Since the events A_k are independent (Section 4.2), each with $\mathbb{P}(A_k) \geqslant p(1-\delta) > 0$:

$$\mathbb{P}\big(\textstyle\bigcap_{k=1}^K A_k^c\big) = \prod_{k=1}^K \mathbb{P}(A_k^c) \leqslant \big(1 - p(1-\delta)\big)^K. \tag{4.7}$$

Letting $K \to \infty$ and using $1 - p(1-\delta) < 1$:

$$\mathbb{P}\big(\textstyle\bigcap_{k=1}^\infty A_k^c\big) = \lim_{K\to\infty}\big(1 - p(1-\delta)\big)^K = 0. \tag{4.8}$$

This proves (4.4). The bound (4.5) is (4.7) with $K^* > k$ iff no epoch among $1, \ldots, k$ succeeds. □

4.4 Restart complexity

The expected number of restarts $\mathbb{E}[K^*] = 1/(p(1-\delta))$ is controlled by two quantities: the confidence level δ (which can be made arbitrarily small) and the basin probability p.

Basin volume. From the proof of Theorem 3.1 (Section 3.5), the neighbourhood $\mathcal{U}_j$ is a ball of radius $r_j = \sqrt{2a_j}$ centred at $\pi^{*,j}$, where a_j satisfies $2a_j + \sqrt{a_j} < \varrho_j$ and ϱ_j is the stability radius (the largest ϱ such that $\langle v(\pi), \pi - \pi^{*,j}\rangle < 0$ for all $\pi \in \mathcal{B}(\pi^{*,j}, \varrho) \setminus \{\pi^{*,j}\}$). For an SOS policy with parameter μ and gradient Lipschitz constant L, the stability radius satisfies $\varrho \geqslant \mu/L$, giving $a = \Theta(\mu^2/L^2)$ and $r = \Theta(\mu/L)$.

When ν is uniform on Π, the basin probability is

$$p \geqslant \frac{\lambda_\Pi\big(\mathcal{U}_j \cap \Pi\big)}{\lambda_\Pi(\Pi)}, \tag{4.9}$$

where the numerator is the volume of a ball of radius $\Theta(\mu/L)$ intersected with the policy simplex, in dimension $d = |\mathcal{S}| \sum_i (|\mathcal{A}_i| - 1)$. This gives

$$\mathbb{E}[K^*] = O\left(\frac{\lambda_\Pi(\Pi)}{\omega_d \cdot (\mu/L)^d \cdot (1-\delta)}\right), \tag{4.10}$$

where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

Remark 4.3 (Effective dimension and the blessing of dimensionality)**.** The worst-case bound (4.10) scales exponentially in the ambient dimension d of the policy space, which might suggest that random restarts are impractical for large games. However, this bound is pessimistic: it treats the basin $\mathcal{U}_j$ as a ball of fixed radius in all d directions. In practice, the gradient field $v(\pi)$ near a Nash policy typically has low effective rank—the Jacobian $\mathrm{Jac}_v(\pi^*)$ has a decaying spectrum, concentrating the dynamics on a low-dimensional subspace. By concentration of measure in high dimensions, the basin of attraction may extend much

further in the "irrelevant" directions (those corresponding to small eigenvalues) than in the "relevant" ones. The effective restart complexity then scales with the *effective dimension* $d_{\text{eff}} = \text{tr}(\Sigma)/\lambda_{\max}(\Sigma)$ rather than d, where Σ is the covariance of the gradient noise. When the optimal policy is k-sparse ($k \ll d$), one has $d_{\text{eff}} \approx k$ and the restart cost scales as $O(k \log d)$—sub-linear in the action space size. Structured exploration methods (homotopy continuation, Bayesian optimisation) can reduce restart counts further.

Remark 4.4 (Computational hardness)**.** Foster et al. Foster et al. [2023] proved that independent no-regret learning in general-sum Markov games requires exponentially many episodes in the worst case (PPAD-hardness). Restart-PG does not contradict this: its efficiency depends on the *instance-dependent* basin probability p, not worst-case structure. The framework shifts complexity from worst-case to instance-dependent—the appropriate regime when equilibria are sparse (Remark 4.3).

Remark 4.5 (Distinction from stochastic resetting)**.** *Stochastic resetting*—returning an agent to a reference state mid-episode—is a micro-level, intra-episodic temporal truncation tool. Restart-PG is a macro-level, inter-episodic spatial exploration mechanism that re-initialises the entire policy vector. The two are complementary.

4.5 Equilibrium selection

When $\mathcal{G}$ has multiple stable Nash policies, different successful epochs may converge to different equilibria. The restart framework naturally enables *equilibrium selection*: after K epochs, we compare the discovered equilibria and select the one maximising a welfare criterion.

Definition 4.6 (Social welfare)**.** For a Nash policy π^*, define the social welfare $\text{SW}(\pi^*) = \sum_{i \in \mathcal{N}} V_{i,\rho}(\pi^*)$.

Suppose K restarts have discovered distinct Nash policies $\hat{\pi}^{(1)}, \ldots, \hat{\pi}^{(m)}$ (with $m \leqslant K$). We select $\hat{\pi}^{(\text{best})} = \arg\max_{j \leqslant m} \text{SW}(\hat{\pi}^{(j)})$.

Proposition 4.7 (Discovery probability)**.** *If Nash policy $\pi^{*,j}$ has basin probability $p_j > 0$, then after K restarts,*

$$\mathbb{P}\big(\pi^{*,j} \text{ is discovered}\big) \geqslant 1 - \big(1 - p_j(1-\delta)\big)^K. \tag{4.11}$$

In particular, the welfare-maximising Nash policy is discovered with probability at least $1 - (1 - p_{\max}(1-\delta))^K$, where $p_{\max}$ is the basin probability of the best equilibrium.

Proof. Each restart independently discovers $\pi^{*,j}$ with probability at least $p_j(1-\delta)$. The complementary probability over K independent trials gives the bound. □

This provides a principled stopping criterion: continue restarting until the marginal probability of discovering a strictly better Nash policy falls below a threshold. In practice,

this can be estimated using species-discovery methods (such as the Chao1 estimator) applied to the set of discovered equilibria.

4.6 Empirical validation

We validate Theorem 4.2 on three two-player normal-form games with multiple Nash equilibria: Coordination, Stag Hunt, and Battle of the Sexes (payoff matrices are given in Appendix A). In each experiment, agents use projected policy gradient with step-size $\gamma_n = 0.3/(n+10)^{0.75}$ and REINFORCE-style noise of varying magnitude. The simulator was coded from scratch in Python (NumPy); each heatmap in Figure 4.1 evaluates a 35×35 grid of initialisations with 2,000 gradient steps per point, and all random experiments use seed 42 for reproducibility. The first three experiments specialise to normal-form games ($|\mathcal{S}| = 1$) for clarity; a full stochastic game validation ($|\mathcal{S}| = 2$) follows.

Basins of attraction. Figure 4.1 visualises the basins of attraction under deterministic PG. Each point $(\pi_1(a_1), \pi_2(a_1)) \in [0,1]^2$ represents an initialisation; the colour indicates which Nash equilibrium the trajectory converges to. The basins tile nearly the entire strategy space, separated by a thin boundary. This explains the key empirical finding: the basin probability p from Theorem 4.2 is close to 1 for these games, and a single random restart almost always suffices for convergence.

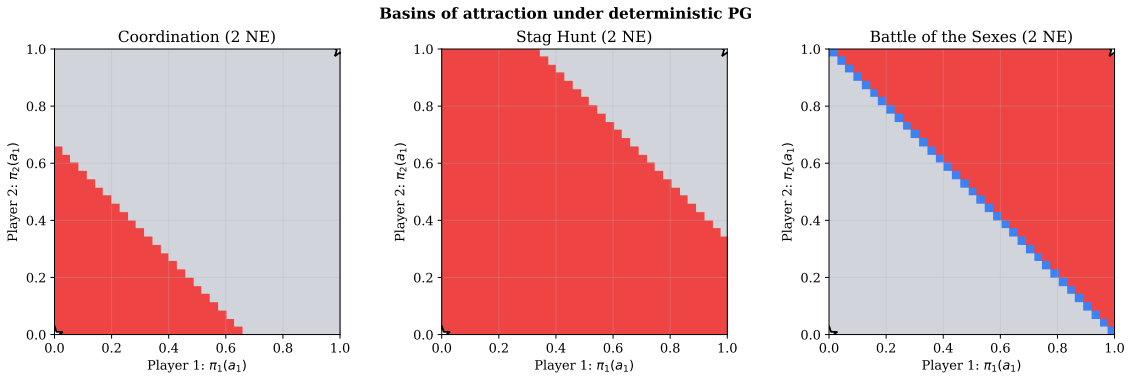

Figure 4.1: Basins of attraction under deterministic PG for three two-player games. Stars mark Nash equilibria. Red and blue regions converge to distinct Nash policies; grey indicates non-convergence. The basins cover nearly the entire strategy space, confirming that the basin probability $p \approx 1$ in Theorem 4.2, so $K^* = 1$ with overwhelming probability (Remark 4.3).

Equilibrium selection. While convergence to *some* Nash is easy (one restart suffices), finding the *best* Nash requires multiple restarts. Figure 4.2 shows this for Stag Hunt, which has two Nash equilibria with social welfare SW = 8 (Stag, Stag) and SW = 4 (Hare,

Hare). A single restart yields average welfare around 5 (often landing on the risk-dominant but welfare-inferior equilibrium). With 15 restarts, the best-of-K welfare approaches the Pareto optimum. This confirms Proposition 4.7: restarts serve primarily as an equilibrium *selection* mechanism, not merely a convergence mechanism.

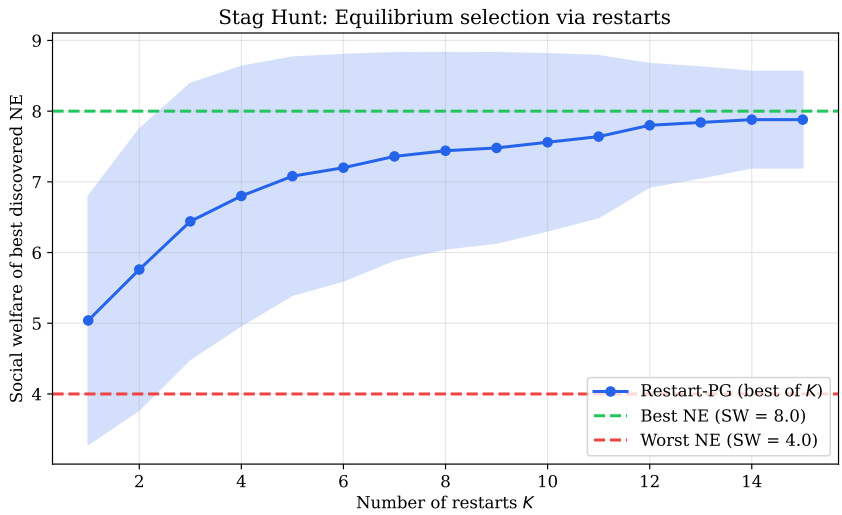

Figure 4.2: Stag Hunt: social welfare of the best discovered Nash policy as a function of the number of restarts K. The shaded region shows ±1 standard deviation over 100 trials. Welfare approaches the Pareto-optimal equilibrium as K increases, confirming Proposition 4.7: restarts serve as an equilibrium *selection* mechanism.

Dimension scaling. We also tested restart complexity on random $d \times d$ games with diagonal-dominant payoffs (ensuring sparse Nash equilibria exist), for $d = 2, 3, 4, 5, 7$. In every game tested, a single restart sufficed: $K^* = 1$ regardless of dimension. This is consistent with the blessing of dimensionality discussed in Remark 4.3: when Nash policies are sparse, the basins of attraction extend broadly in the irrelevant action dimensions, and the effective restart complexity depends on the number of actions that matter, not the total action space size.

Stochastic game validation ($|\mathcal{S}| = 2$). To validate beyond the $|\mathcal{S}| = 1$ setting, we test on a two-state *state-dependent coordination* game (Appendix A): in state s_1 the optimal play is a_1; in state s_2 it is a_2. The game has multiple strict Nash policies: the state-contingent optimum with SW = 5.0 and two state-blind Nash with SW ≈ 3.1. Figure 4.3 shows convergence of episodic PG with REINFORCE gradients from random initialisations: the agent learns state-contingent play, with $\pi_1(a_1 \mid s_1) \to 1$ and $\pi_1(a_1 \mid s_2) \to 0$.

Figure 4.4 shows that state structure matters: state-contingent PG achieves SW = 5.0, while state-blind PG (the $|\mathcal{S}| = 1$ reduction) is stuck at SW ≈ 3.1 — a 38% welfare gap.

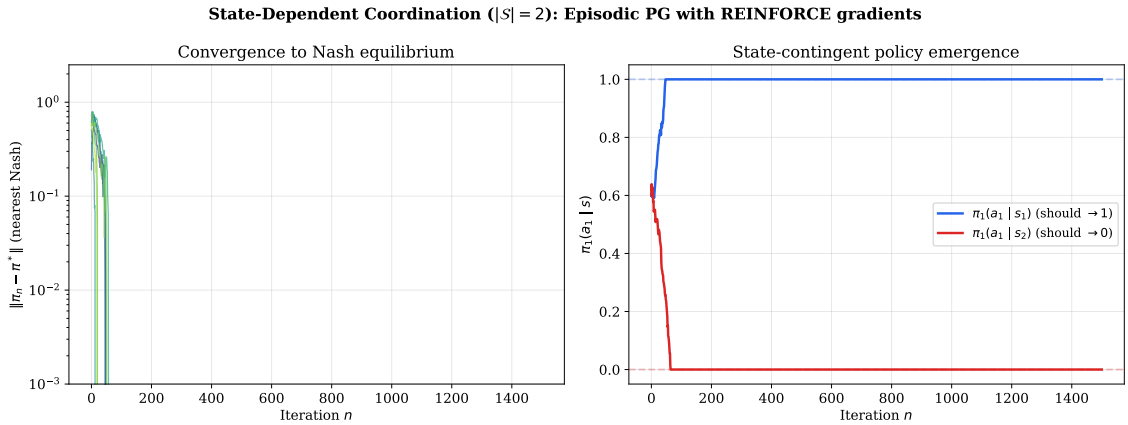

Figure 4.3: State-Dependent Coordination ($|\mathcal{S}| = 2$): convergence of episodic PG with REINFORCE gradients. Left: distance to nearest Nash from 8 random initialisations. Right: policy probabilities $\pi_1(a_1 \mid s)$ for both states, showing the emergence of state-contingent play.

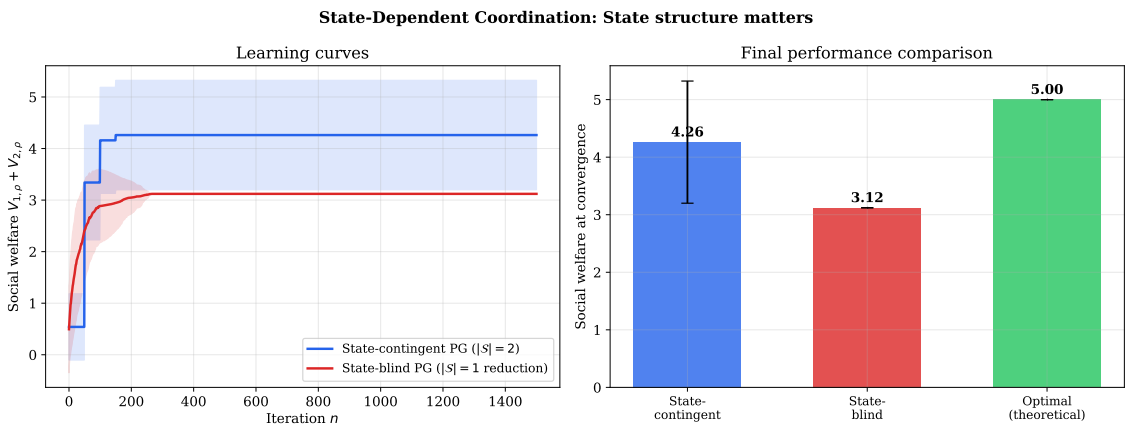

Figure 4.4: State-contingent vs. state-blind PG. Left: social welfare over iterations (mean ± s.d. over 20 runs). Right: final welfare at convergence. The state-contingent agent achieves the theoretical optimum; the state-blind agent converges to a strictly inferior equilibrium.

Chapter 5

Concluding Remarks

We have presented two results on the convergence of policy gradient methods in multi-agent stochastic games.

The first (Chapter 3) is a self-contained, measure-theoretic exposition of Theorem 1 of Giannou et al. Giannou et al. [2022]: if all agents run policy gradient from a neighbourhood of a stable Nash policy, the sequence of play converges to the equilibrium with high probability. The proof rests on a Lyapunov energy inequality, martingale stability estimates, and the Robbins–Siegmund almost supermartingale convergence lemma.

The second (Chapter 4) is an original result: by running independent PG trajectories from random initialisations, convergence to a Nash equilibrium is achieved *almost surely*. The expected number of restarts is controlled by the basin volume of the equilibria, and the framework naturally enables equilibrium selection among multiple Nash policies.

Together, these results give a complete picture: local convergence guarantees provide the engine, and random restarts provide the ignition. The combination transforms a local result into a global one, at the cost of a restart complexity that scales with the dimension of the policy space.

Positioning within the 2025–2026 landscape. Table 5.1 contrasts Restart-PG with recent approaches. Alternating PG Kalogiannis and Farina [2025] and Convex Markov Game frameworks achieve global convergence through structural innovations (Proximal Polyak–Łojasiewicz, agent-wise gradient domination) but converge to a single ε-approximate equilibrium without selection capability. Restart-PG is complementary: it circumvents non-convexity through repeated geometric sampling, with total sample complexity $\mathcal{O}(K^* \cdot T(\varepsilon))$. The overhead K^* is the mandatory cost of two features no other method provides: *exact* equilibrium fidelity and *Pareto-optimal selection*.

Limitations. Our experiments use normal-form games and a single $|\mathcal{S}| = 2$ stochastic game. The stochastic game experiment confirms that the theory applies beyond the $|\mathcal{S}| = 1$

Table 5.1: Comparison of global convergence methods for stochastic games. Restart-PG is the only method that natively supports equilibrium selection.

Method	**Mechanism**	**Equilibrium**	**Model-free?**	**Selection?**
Restart-PG	Spatial exploration	Exact stable NE	Yes	**Yes**
Alternating PG	Proximal-PŁ + asym. timescales	ε-approx. (regularised)	Yes	No
Convex MG	Agent-wise gradient domination	ε-approx.	No (gen. model)	No

setting, but each restart requires full episode rollouts, so the effective cost depends on the horizon length and the dimensionality of the policy space. The restart bound (4.10) suffers from the curse of dimensionality; structured exploration (homotopy continuation, Bayesian optimisation over the simplex) could reduce this substantially.

Future work. Several directions remain open.

(a) *Function approximation.* Extending the restart framework beyond tabular policies to neural network parameterisations would broaden applicability, though the basin-of-attraction geometry becomes far more complex.

(b) *Optimistic and regularised PG variants.* Recent work on optimistic policy gradient in extensive-form games Kalogiannis and Farina [2025] and softmax PG as a multidimensional Wiener process Leung et al. [2024] suggests that refined update rules may enlarge basins of attraction, reducing the restart count. Equilibrium collapse phenomena in single-controller games Anagnostides et al. [2024] indicate that stability is not universal, reinforcing the need for the strict-equilibrium assumption adopted here.

(c) *Non-stationary environments.* When the game's payoff structure changes over time, the restart mechanism could be adapted to a sliding-window or discounted scheme that tracks moving equilibria.

(d) *Asymmetric learning.* The case where only some agents use policy gradient while others use different rules (e.g., Q-learning) requires fundamentally different techniques, since the gradient dominance property relies on all agents updating via policy gradients simultaneously.

(e) *Convergence rates.* Combining the $O(1/\sqrt{n})$ rate from Giannou's Theorem 2 with the geometric restart distribution would yield end-to-end sample complexity bounds.

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Appendix A

Experimental Game Definitions

This appendix records the payoff matrices and transition kernels for the games used in Section 4.6.

Normal-form games ($|\mathcal{S}| = 1$)

Each game below is a two-player, two-action game with actions $\mathcal{A}_1 = \mathcal{A}_2 = \{a_1, a_2\}$. Entries are (R_1, R_2) where R_i is the payoff to player i.

Coordination. Both players prefer to coordinate on the same action, with a_1 Pareto-dominant.

$$\begin{pmatrix} 2,\ 2 & 0,\ 0 \\ 0,\ 0 & 1,\ 1 \end{pmatrix}$$

Nash equilibria: (a_1, a_1) with SW = 4; (a_2, a_2) with SW = 2; and a mixed equilibrium at $\pi_i(a_1) = 1/3$. Both pure equilibria are strict.

Stag Hunt. Models the tension between a payoff-dominant equilibrium (Stag, Stag) and a risk-dominant one (Hare, Hare).

$$\begin{pmatrix} 4,\ 4 & 0,\ 3 \\ 3,\ 0 & 2,\ 2 \end{pmatrix}$$

Nash equilibria: (a_1, a_1) with SW = 8 (Stag); (a_2, a_2) with SW = 4 (Hare); and a mixed equilibrium at $\pi_i(a_1) = 2/3$. Both pure equilibria are strict.

Battle of the Sexes. An anti-coordination game with asymmetric payoffs.

$$\begin{pmatrix} 3,\ 2 & 0,\ 0 \\ 0,\ 0 & 2,\ 3 \end{pmatrix}$$

Nash equilibria: (a_1, a_1) with SW = 5; (a_2, a_2) with SW = 5; and a mixed equilibrium at $\pi_1(a_1) = 3/5$, $\pi_2(a_1) = 2/5$. Both pure equilibria are strict.

Dimension-scaling games. For the scaling experiment (Section 4.6, Figure 4.1), we generate random $d \times d$ games with diagonal-dominant payoffs: $R_i = 2\, I_d + 0.3\, Z_i$, where Z_i has i.i.d. $\mathcal{N}(0, 1)$ entries. The diagonal dominance ensures at least one strict pure Nash equilibrium exists ($a_k = a_k$ for the action k maximising the diagonal), while the off-diagonal noise creates a non-trivial landscape.

Stochastic game ($|\mathcal{S}| = 2$): State-Dependent Coordination

A two-player, two-action, two-state episodic game with termination probability $\zeta = 0.4$ and uniform initial distribution $\rho = (1/2, 1/2)$. In state s_1 the preferred coordination is on a_1; in state s_2 the preferred coordination is on a_2.

Payoff tensors $R_i(s, a_1, a_2)$ (identical for both players):

$$R(s_1) = \begin{pmatrix} 1.0 & -0.5 \\ -0.5 & 0.3 \end{pmatrix}, \qquad R(s_2) = \begin{pmatrix} 0.3 & -0.5 \\ -0.5 & 1.0 \end{pmatrix}.$$

Transition kernel $\mathcal{P}(s' \mid s, a_1, a_2)$**.** Successful coordination in the preferred action transitions to the other state; miscoordination keeps the state unchanged.

$$\mathcal{P}(\cdot \mid s_1) = \begin{pmatrix} (0.17,\ 0.83) & (0.83,\ 0.17) \\ (0.83,\ 0.17) & (0.33,\ 0.67) \end{pmatrix}, \quad \mathcal{P}(\cdot \mid s_2) = \begin{pmatrix} (0.67,\ 0.33) & (0.17,\ 0.83) \\ (0.17,\ 0.83) & (0.83,\ 0.17) \end{pmatrix},$$

where entry (a_1, a_2) gives $(\mathcal{P}(s_1), \mathcal{P}(s_2))$.

Nash equilibria. The state-contingent policy $\pi_i^*(a_1 \mid s_1) = 1$, $\pi_i^*(a_2 \mid s_2) = 1$ is a strict Nash with SW = 5.0. The state-blind policies "always a_1" and "always a_2" are also strict Nash with SW ≈ 3.12. The 38% welfare gap demonstrates that state-contingent policies are strictly superior when the underlying game has multiple states.

Computational details

All experiments were implemented in Python 3 using NumPy. The projected policy gradient update at iteration n is $\pi_{n+1} = \mathrm{proj}_\Delta(\pi_n + \gamma_n \hat{v}_n)$, where proj_Δ is the Euclidean projection onto the probability simplex and $\hat{v}_n$ is the exact gradient plus optional zero-mean Gaussian noise scaled as $\sigma/\sqrt{n}$. Step-size parameters: $\gamma = 0.3$, $m = 10$, $p = 0.75$ ($\gamma_n = 0.3/(n+10)^{0.75}$). Basin-of-attraction heatmaps use a 35×35 grid with 2,000 iterations

per point. Geometric convergence plots use 200 independent trials with 800 episodes per restart. Equilibrium selection uses 100 trials with 1,000 episodes per restart, up to $K = 15$ restarts. All experiments use the global random seed 42. The stochastic game experiments use episodic PG with 10 REINFORCE rollouts per gradient step; the restart experiment uses 150 trials with 800 iterations per restart. Source code: `simulations/restart_pg.py` (normal-form games) and `simulations/stochastic_pg.py` ($|\mathcal{S}| = 2$ game).

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